MultiOutput K-Nearest Neighbors (KNN) with PCA Model: Yearly Analysis and Predictions

```
!pip install plotly
import pandas as pd
pd.options.mode.chained_assignment = None
```

```
Requirement already satisfied: plotly in /opt/conda/lib/python3.10/site-packages (5.24.1)
Requirement already satisfied: tenacity>=6.2.0 in /opt/conda/lib/python3.10/site-packages (from plotly) (9.0.0)
Requirement already satisfied: packaging in /opt/conda/lib/python3.10/site-packages (from plotly) (22.0)
```

MultiOutput KNN model of Pharmasuticals Data of 2014

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsRegressor
from sklearn.multioutput import MultiOutputRegressor
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score, f1_score, confusion_matrix, recall_scor
from sklearn.preprocessing import StandardScaler, LabelEncoder
import numpy as np

# Load dataset
df = pd.read_csv("meps_clean_2014.csv")

# Define features (X) and target variables (y)
X = df[['Age', 'Sex', 'Ethnicity', 'Race', 'Personal_income', 'Family_income',
        'Poverty_category', 'Insurance_coverage', 'PMED_insurance_coverage', 'Can\'t_get_PMED']]
y = df[['Concentration', 'Drug']]

feature_set = ['Drug', 'Concentration', 'Age', 'Sex', 'Ethnicity',
               'Race', 'Personal_income', 'Family_income',
               'Poverty_category', 'Insurance_coverage',
               'PMED_insurance_coverage', 'Can\'t_get_PMED']

# Drop rows with missing values in any of the features or target variables
df_cleaned = df.dropna(subset=feature_set)

# Handle categorical columns in X (features) and convert them to numeric using LabelEncoder
categorical_columns = X.select_dtypes(include=['object']).columns
for col in categorical_columns:
    X[col] = LabelEncoder().fit_transform(X[col])

# Handle missing values in target variables y (Concentration and Drug)
encoder = LabelEncoder()
y['Drug'] = encoder.fit_transform(y['Drug'])

# Replace infinite values with NaN
y.replace([np.inf, -np.inf], np.nan, inplace=True)

# Convert Concentration to numeric values and handle any NaN values
y['Concentration'] = pd.to_numeric(y['Concentration'], errors='coerce')

# Ensure there are no missing values in features or targets after conversion
X = X.fillna(0)
y = y.fillna(0)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Standardize the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Initialize the KNN Regressor
knn_model = KNeighborsRegressor(n_neighbors=5)
multi_output_knn = MultiOutputRegressor(knn_model)

# Train the model
multi_output_knn.fit(X_train_scaled, y_train)

# Make predictions
y_pred = multi_output_knn.predict(X_test_scaled)

# Evaluate Concentration (regression)
mse = mean_squared_error(y_test['Concentration'], y_pred[:, 0])
r2 = r2_score(y_test['Concentration'], y_pred[:, 0])
```

```
# Print metrics for Concentration
print("\nRegression Metrics for Concentration:")
print(f"Mean Squared Error (MSE): {mse}")
print(f"R² Score: {r2}")

# Evaluate Drug (classification)
threshold = 0.5
y_test_binarized = (y_test['Drug'] >= threshold).astype(int)
y_pred_binarized = (y_pred[:, 1] >= threshold).astype(int)

conf_matrix = confusion_matrix(y_test_binarized, y_pred_binarized)
accuracy = accuracy_score(y_test_binarized, y_pred_binarized)
f1 = f1_score(y_test_binarized, y_pred_binarized)
recall = recall_score(y_test_binarized, y_pred_binarized)

# Print metrics for Drug
print("\nClassification Metrics for Drug:")
print(f"Accuracy: {accuracy}")
print(f"F1 Score: {f1}")
print(f"Recall: {recall}")
print("\nConfusion Matrix:")
print(conf_matrix)

# Create a DataFrame to compare actual vs predicted values
comparison_df = pd.DataFrame({
    'Actual_Concentration': y_test['Concentration'],
    'Predicted_Concentration': y_pred[:, 0],
    'Actual_Drug': y_test['Drug'],
    'Predicted_Drug': y_pred[:, 1]
})

# Add inverse-transformed Drug names
comparison_df['Predicted_Drug_Name'] = encoder.inverse_transform(comparison_df['Predicted_Drug'].round().astype(i
comparison_df['Actual_Drug_Name'] = encoder.inverse_transform(comparison_df['Actual_Drug'].round().astype(int))

# Save results to CSV
comparison_df.to_csv("comparison_results_2014.csv", index=False)

# Display comparison table
print("\nComparison of Actual vs Predicted Values:")
print(comparison_df.head())

# Ensure 'Drug' is numeric and included in the cleaned dataset
df_cleaned['Drug'] = encoder.transform(df_cleaned['Drug'])

# Correlation matrix and heatmap (including Drug)
correlation_matrix = df_cleaned[feature_set].corr()

# Plot the heatmap
plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5,
            cbar_kws={'label': 'Correlation Coefficient'})
plt.title("Correlation Heatmap Including Drug")
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

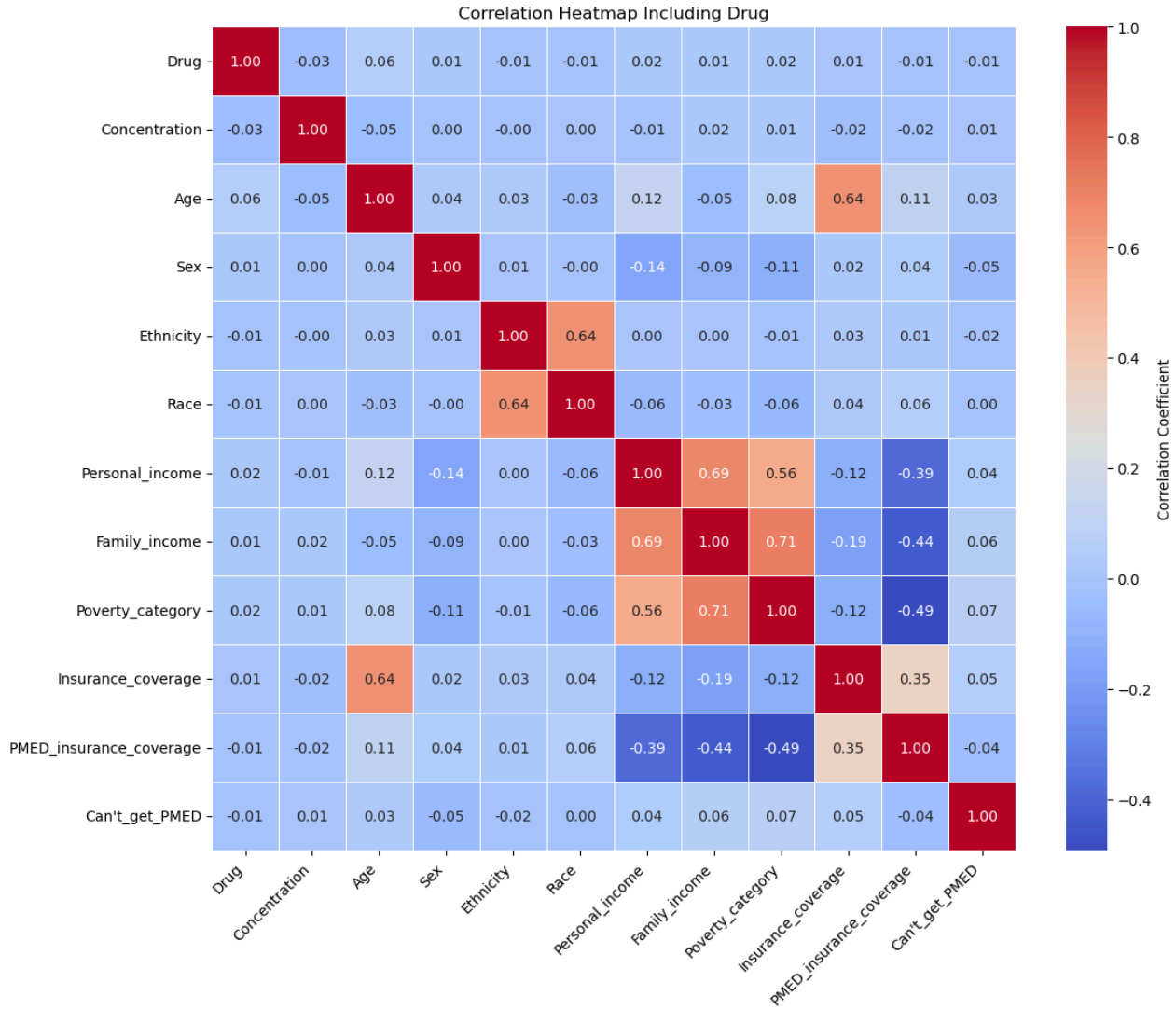
```
Regression Metrics for Concentration:
Mean Squared Error (MSE): 1.0988431161012617
R² Score: 0.13494099933483272

Classification Metrics for Drug:
Accuracy: 0.9932475292095517
F1 Score: 0.996612118349999
Recall: 0.9999175971323802

Confusion Matrix:
[[    3   326]
 [    4 48538]]

Comparison of Actual vs Predicted Values:
        Actual_Concentration  Predicted_Concentration  Actual_Drug  \
217240              0.000024                 0.000853          145
216189              0.000739                 0.000582          274
179156              0.000068                 0.000068          130
192999              0.001450                 0.001693          334
220253              0.000008                 0.000045          113

        Predicted_Drug Predicted_Drug_Name Actual_Drug_Name
217240           162.2      INSULIN LISPRO       GUANFACINE
216189           270.6         PROPAFENONE       QUETIAPINE
179156           188.6          LOPERAMIDE     FLUOCINONIDE
192999           194.0          LOVASTATIN        VERAPAMIL
220253           146.2         HALOPERIDOL        ESTRADIOL
```



Runing by reducing the features

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsRegressor
from sklearn.multioutput import MultiOutputRegressor
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score, f1_score, confusion_matrix, recall_sco
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.impute import SimpleImputer
import numpy as np

# Load dataset
df = pd.read_csv("meps_clean_2014.csv")

# Define features (X) and target variables (y)
X = df[['Age', 'Sex', 'Ethnicity', 'Race', 'Personal_income', 'Family_income',
        'Poverty_category', 'Insurance_coverage', 'Can\'t_get_PMED']]
y = df[['Concentration', 'Drug']]

# Drop rows with missing values in any of the features or target variables (if any)
df_cleaned = df.dropna(subset=feature_set)

# Handle categorical columns in X (features) and convert them to numeric using LabelEncoder
categorical_columns = X.select_dtypes(include=['object']).columns
for col in categorical_columns:
    X[col] = LabelEncoder().fit_transform(X[col])

# Handle missing values in target variables y (Concentration and Drug)
encoder = LabelEncoder()
y['Drug'] = encoder.fit_transform(y['Drug'])

# Replace infinite values with NaN in target variables
y.replace([np.inf, -np.inf], np.nan, inplace=True)

# Convert Concentration to numeric values and handle any NaN values
y['Concentration'] = pd.to_numeric(y['Concentration'], errors='coerce')

# Use SimpleImputer to handle any NaN values in the features (X) and target (y)
imputer = SimpleImputer(strategy='mean')  # Replace NaN with the mean of the column
X = pd.DataFrame(imputer.fit_transform(X), columns=X.columns)
y = pd.DataFrame(imputer.fit_transform(y), columns=y.columns)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Standardize the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Initialize the KNN Regressor
knn_model = KNeighborsRegressor(n_neighbors=5)
multi_output_knn = MultiOutputRegressor(knn_model)

# Train the model
multi_output_knn.fit(X_train_scaled, y_train)

# Make predictions
y_pred = multi_output_knn.predict(X_test_scaled)

# Evaluate Concentration (regression)
mse = mean_squared_error(y_test['Concentration'], y_pred[:, 0])
r2 = r2_score(y_test['Concentration'], y_pred[:, 0])

# Print metrics for Concentration
print("\nRegression Metrics for Concentration:")
print(f"Mean Squared Error (MSE): {mse}")
print(f"R² Score: {r2}")

# Evaluate Drug (classification)
threshold = 0.5
y_test_binarized = (y_test['Drug'] >= threshold).astype(int)
y_pred_binarized = (y_pred[:, 1] >= threshold).astype(int)

conf_matrix = confusion_matrix(y_test_binarized, y_pred_binarized)
accuracy = accuracy_score(y_test_binarized, y_pred_binarized)
f1 = f1_score(y_test_binarized, y_pred_binarized)
recall = recall_score(y_test_binarized, y_pred_binarized)

# Print metrics for Drug
print("\nClassification Metrics for Drug:")
print(f"Accuracy: {accuracy}")
```

```
print(f"F1 Score: {f1}")
print(f"Recall: {recall}")
print("\nConfusion Matrix:")
print(conf_matrix)

# Create a DataFrame to compare actual vs predicted values
comparison_df = pd.DataFrame({
    'Actual_Concentration': y_test['Concentration'],
    'Predicted_Concentration': y_pred[:, 0],
    'Actual_Drug': y_test['Drug'],
    'Predicted_Drug': y_pred[:, 1]
})

# Add inverse-transformed Drug names
comparison_df['Predicted_Drug_Name'] = encoder.inverse_transform(comparison_df['Predicted_Drug'].round().astype(
comparison_df['Actual_Drug_Name'] = encoder.inverse_transform(comparison_df['Actual_Drug'].round().astype(int))

# Save results to CSV
comparison_df.to_csv("comparison_results_2014.csv", index=False)

# Display comparison table
print("\nComparison of Actual vs Predicted Values:")
print(comparison_df.head())

# Ensure 'Drug' is numeric and included in the cleaned dataset
df_cleaned['Drug'] = encoder.transform(df_cleaned['Drug'])

# Correlation matrix and heatmap (including Drug)
correlation_matrix = df_cleaned[feature_set].corr()

# Plot the heatmap
plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5,
            cbar_kws={'label': 'Correlation Coefficient'})
plt.title("Correlation Heatmap Including Drug")
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

```
Regression Metrics for Concentration:
Mean Squared Error (MSE): 1.0891264563408305
R² Score: 0.14259033500104934

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192999           194.0          LOVASTATIN        VERAPAMIL
220253           145.4          GUANFACINE        ESTRADIOL
```
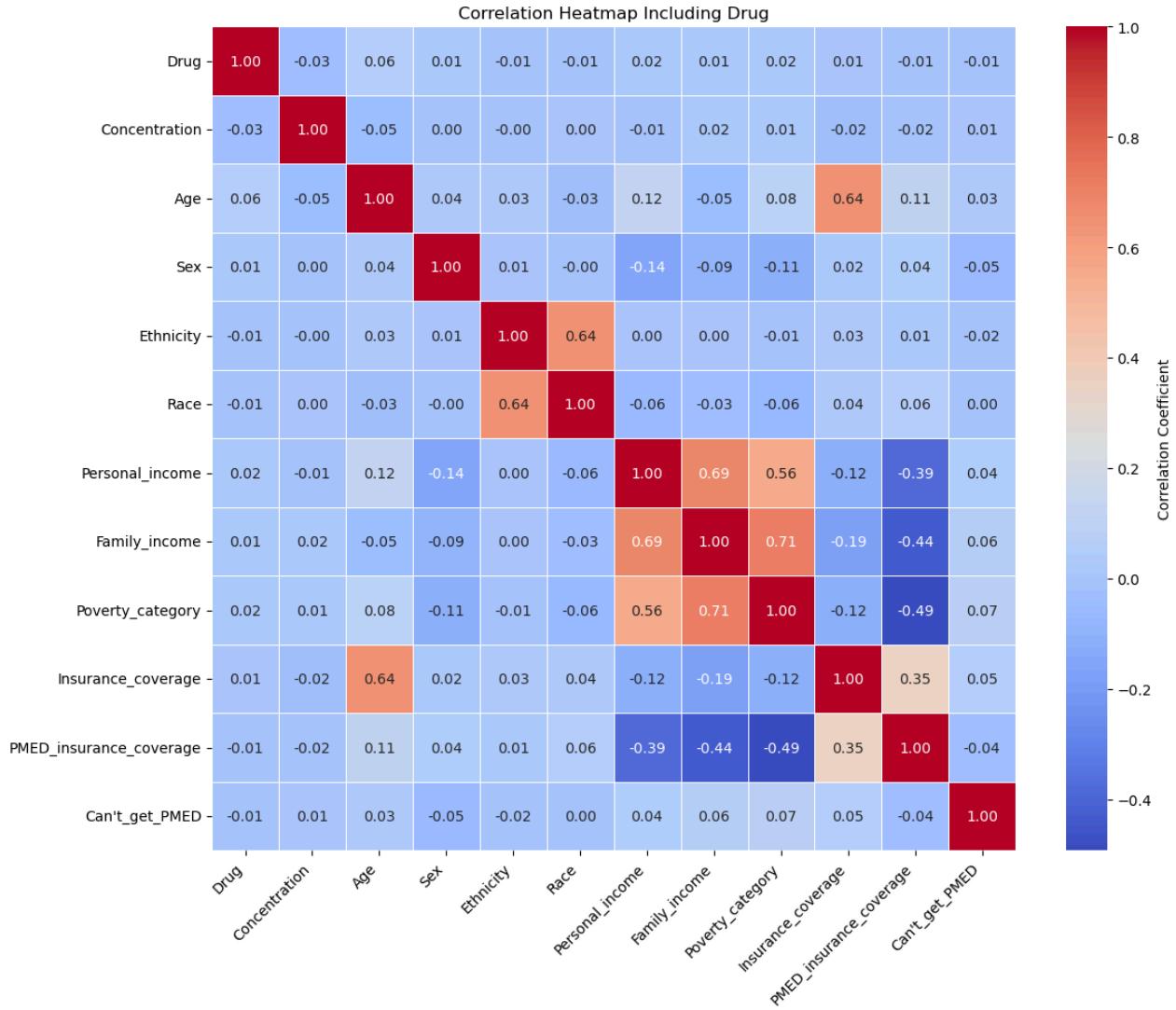



Analysing:

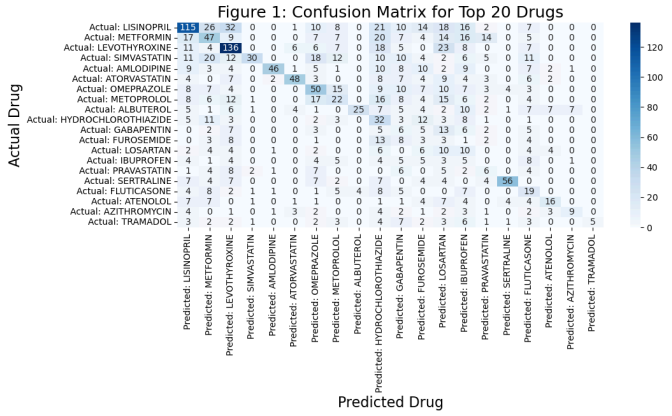
Upon removing the `PMED_insurance_coverage` attribute, the model's efficiency in evaluating `Concentration` improved. This was evident as the **MSE** marginally decreased and the R^2 value increased. However, the **F1 Score**, **Accuracy**, and **Precision** remained unchanged.

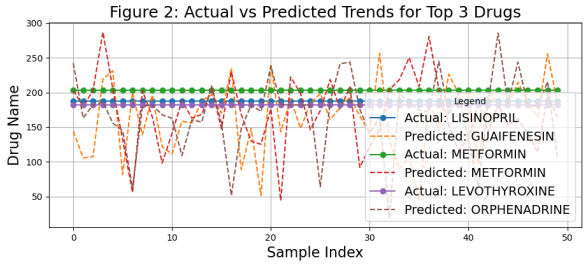
Visualization for Drug Classification

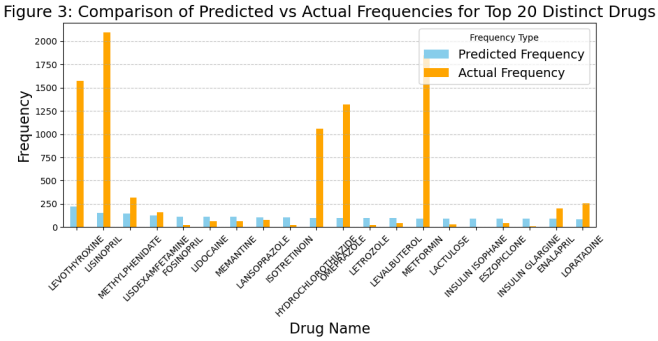
Confusion Matrix, Actual vs. Predicted Trends (Line Plot), and Predicted vs. Actual Frequencies (Bar Plot)

```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix
import numpy as np
import pandas as pd

# Assuming `df_cleaned`, `y_test`, `y_pred`, and `encoder` are already defined.

# Create a 2x2 grid for the plots, but only use 3 spaces with larger figsize
fig, axes = plt.subplots(2, 2, figsize=(24, 18))  # Increased the figsize to make all plots larger

# ---- Common Block to Calculate Top Drugs ----
top_n = 20  # You can change this to 10, 5, or any other number as required
top_drugs = df_cleaned['Drug'].value_counts().nlargest(top_n).index

# ---- Figure 1: Confusion Matrix for Top Drugs ----
filtered_y_test = y_test[y_test['Drug'].isin(top_drugs)]['Drug']
filtered_y_pred = y_pred[:, 1][np.isin(y_test['Drug'], top_drugs)].round()
conf_matrix = confusion_matrix(filtered_y_test, filtered_y_pred, labels=top_drugs)

sns.heatmap(
    conf_matrix,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=[f"Predicted: {encoder.inverse_transform([int(label)])[0]}" for label in top_drugs],
    yticklabels=[f"Actual: {encoder.inverse_transform([int(label)])[0]}" for label in top_drugs],
    ax=axes[0, 0]
)
axes[0, 0].set_title(f"Figure 1: Confusion Matrix for Top {top_n} Drugs", fontsize=18)
axes[0, 0].set_xlabel("Predicted Drug", fontsize=16)
axes[0, 0].set_ylabel("Actual Drug", fontsize=16)

# ---- Figure 2: Actual vs Predicted Trends for Top Drugs ----
for drug in top_drugs[:3]:  # Limiting to the top 5 for this graph
    actual_indices = y_test['Drug'] == drug
    actual = y_test.loc[actual_indices, 'Drug'].head(50).values
    predicted = y_pred[actual_indices, 1][:50]
    if len(actual) > 0:
        axes[0, 1].plot(
            range(len(actual)),
            actual,
            label=f"Actual: {encoder.inverse_transform([drug])[0]}",
            marker='o'
        )
        axes[0, 1].plot(
            range(len(predicted)),
            predicted,
            label=f"Predicted: {encoder.inverse_transform([int(predicted[0])])[0]}",
            linestyle='--'
        )
axes[0, 1].set_title(f"Figure 2: Actual vs Predicted Trends for Top 3 Drugs", fontsize=18)
axes[0, 1].set_xlabel("Sample Index", fontsize=16)
axes[0, 1].set_ylabel("Drug Name", fontsize=16)
axes[0, 1].legend(title="Legend", fontsize=14)
axes[0, 1].grid()

# ---- Figure 3: Bar Plot for Top Drugs ----
top_predicted_drugs = pd.Series(y_pred[:, 1]).value_counts().sort_values(ascending=False)
top_actual_frequencies = y_test['Drug'].value_counts()
common_drugs = top_predicted_drugs.index.intersection(top_actual_frequencies.index)

distinct_predicted_drugs = top_predicted_drugs.loc[common_drugs].head(top_n)
distinct_actual_frequencies = top_actual_frequencies.loc[common_drugs].head(top_n)

# Decode drug names using the LabelEncoder
decoded_drug_names = encoder.inverse_transform(distinct_predicted_drugs.index.astype(int))

# Create a comparison DataFrame for distinct drugs
comparison_df = pd.DataFrame({
```

```
    'Drug': decoded_drug_names,  # Use drug names for better readability
    'Predicted Frequency': distinct_predicted_drugs.values,
    'Actual Frequency': distinct_actual_frequencies.values
})

# Plot the comparison bar chart with a larger size
comparison_df.set_index('Drug').plot(kind='bar', figsize=(20, 12), rot=45, color=['skyblue', 'orange'], ax=axes[
axes[1, 0].set_title(f"Figure 3: Comparison of Predicted vs Actual Frequencies for Top {top_n} Distinct Drugs",
axes[1, 0].set_xlabel("Drug Name", fontsize=16)
axes[1, 0].set_ylabel("Frequency", fontsize=16)
axes[1, 0].legend(title="Frequency Type", loc='upper right', fontsize=14)
axes[1, 0].grid(axis='y', linestyle='--', alpha=0.7)

# Remove the extra plot (bottom-right corner) to leave only 3 plots
fig.delaxes(axes[1, 1])

# Adjust layout and show the plots
plt.tight_layout()
plt.show()
```







Analysis:

Figure 1: Confusion Matrix

The confusion matrix for the top 10 predicted drugs, shows that the model well performed in classifying the drugs. For common drugs such as **Levothyroxine, Metformin, and Lisinopril** its more likely to be correctly predicted, with fewer misclassifications while such as **Amoxicillin** and **Simvastatin**, have some off-diagonal elements suggesting the model occasionally predicts them incorrectly for drugs like **Hydrochlorothiazide** and **Amlodipine**. **This implies that the model is more accurate for frequently appearing drugs but struggles with less common or similarly labeled drugs.**

Figure 2: Line Plot: Actual vs Predicted

The next plot that for each of the drugs - Metformin, Levothyroxine, we see distinct trends in both actual and predicted values. For Metformin show a relatively close match between actual and predicted values, suggesting that the model performs well for these drugs. However, other drugs like Levothyroxine show a slight divergence between actual and predicted values, this suggests that the

model might not be capturing all the features in the data for those particular drugs, possibly due to complex or less frequent interactions. Despite this, the overall model's performance for the most common drugs indicates a solid prediction ability.

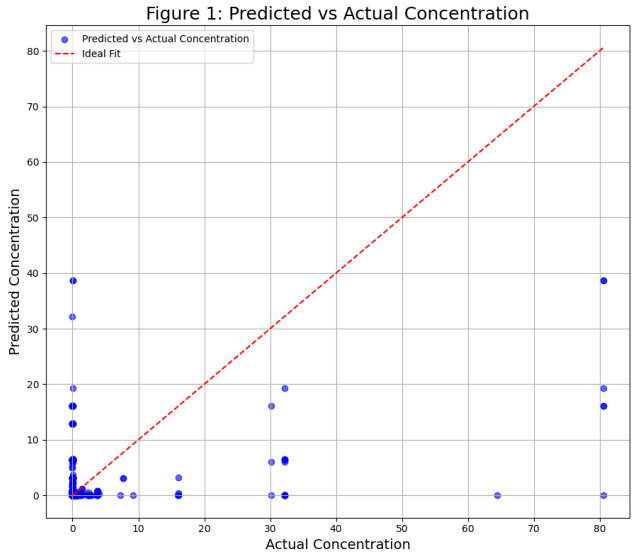
Figure 3: Bar Plot: Predicted vs Actual Frequencies

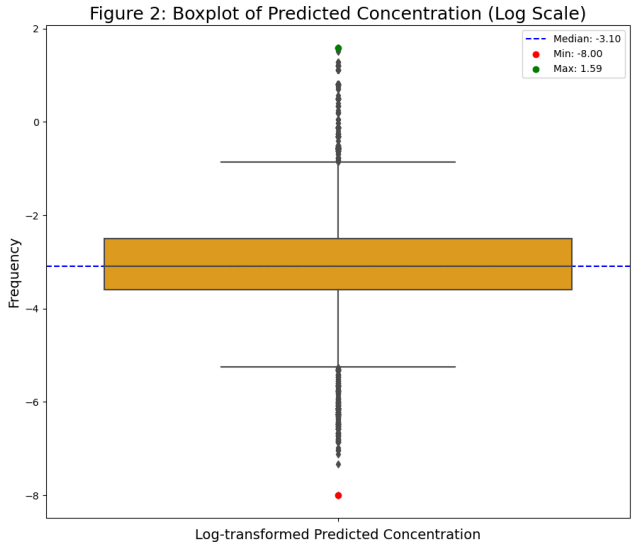
This bar plot shows a significant differences between the predicted and actual frequencies for some drug. For instance, drugs like **Amlodipine** and **Simvastatin** show very close alignment between the predicted and actual frequencies, showing the model is **able to handle commonly encountered drugs quite accurately**. However, drugs like **Hydrochlorothiazide** and **Levothyroxine** show discrepancies, with predicted frequencies much higher than the actual ones, possibly reflecting an overestimation by the model for these particular drugs this shows for rarer drugs the model tends to **over-predict for certain drugs**, possibily due to the class imbalance or insufficient representation of rare drug classes in the dataset.

Visualization For Concentration Regression

Predicted vs Actual for Concentration

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

fig, axs = plt.subplots(1, 2, figsize=(18, 8))

# --- Figure 1: Scatter plot, Predicted vs Actual Concentration ---

axs[0].scatter(y_test['Concentration'], y_pred[:, 0], alpha=0.6, color='blue', label='Predicted vs Actual Concen
axs[0].plot([min(y_test['Concentration']), max(y_test['Concentration'])],
            [min(y_test['Concentration']), max(y_test['Concentration'])],
            color='red', linestyle='--', label='Ideal Fit')

axs[0].set_title("Figure 1: Predicted vs Actual Concentration", fontsize=18)
axs[0].set_xlabel("Actual Concentration", fontsize=14)
axs[0].set_ylabel("Predicted Concentration", fontsize=14)
axs[0].legend()
axs[0].grid()


# --- Figure 2: Boxplot of Log-transformed Predicted Concentration ---

# Shift the values and apply log transformation to predicted concentration
shifted_concentration = y_pred[:, 0] - np.min(y_pred[:, 0]) + 1e-8
log_predicted_concentration = np.log10(shifted_concentration)

sns.boxplot(y=log_predicted_concentration, color='orange', ax=axs[1])

median_value = np.median(log_predicted_concentration)
axs[1].axhline(median_value, color='blue', linestyle='--', label=f'Median: {median_value:.2f}')
min_value = np.min(log_predicted_concentration)
max_value = np.max(log_predicted_concentration)
axs[1].scatter([0], [min_value], color='red', zorder=5, label=f'Min: {min_value:.2f}')
axs[1].scatter([0], [max_value], color='green', zorder=5, label=f'Max: {max_value:.2f}')
axs[1].set_title("Figure 2: Boxplot of Predicted Concentration (Log Scale)", fontsize=18)
axs[1].set_xlabel("Log-transformed Predicted Concentration", fontsize=14)
axs[1].set_ylabel("Frequency", fontsize=14)
axs[1].legend(loc="upper right")

plt.tight_layout()
plt.show()
```





Analysis of Predicted vs. Actual Concentration:

Figure 1:

From the scattere plot we can see there is a weak corelation btween the predicted values and the actual values (correlation coefficient = `0.38`).It can be seen that many predictions diverge from the `ideal fit` (represented by the red dotted line), prominantly at the higher concentration levels. There are many prominant outliers and the grpah is very scattered so this clearly indicats that the model is not working well for the Concentration.

Figure 2:

From the boxplot we can see the **median predicted concentration** to be around **-3**, so half of the predictions fall below this value. While the boxplot ranges from **-4 and -2** so most of the predictions are clustered between these values, that means the model frequently predicts **low concentrations** in this range. The **minimum value** is **-8** indicating outliers while the **maximum value** is **1.59** this shows that the rearely model predicts higher concentrations. The **whiskers** range from **-2** to **-5** showing the majority of the higher values and not predicted correctly and are underpredicted while the model is more leaning towards predicting the lower concentrations.

1. Visualize Predictions for Drug and Concentration Visualize Predictions vs. Actual Values for Drug

Visualize Predictions vs. Actual Values for Concentration

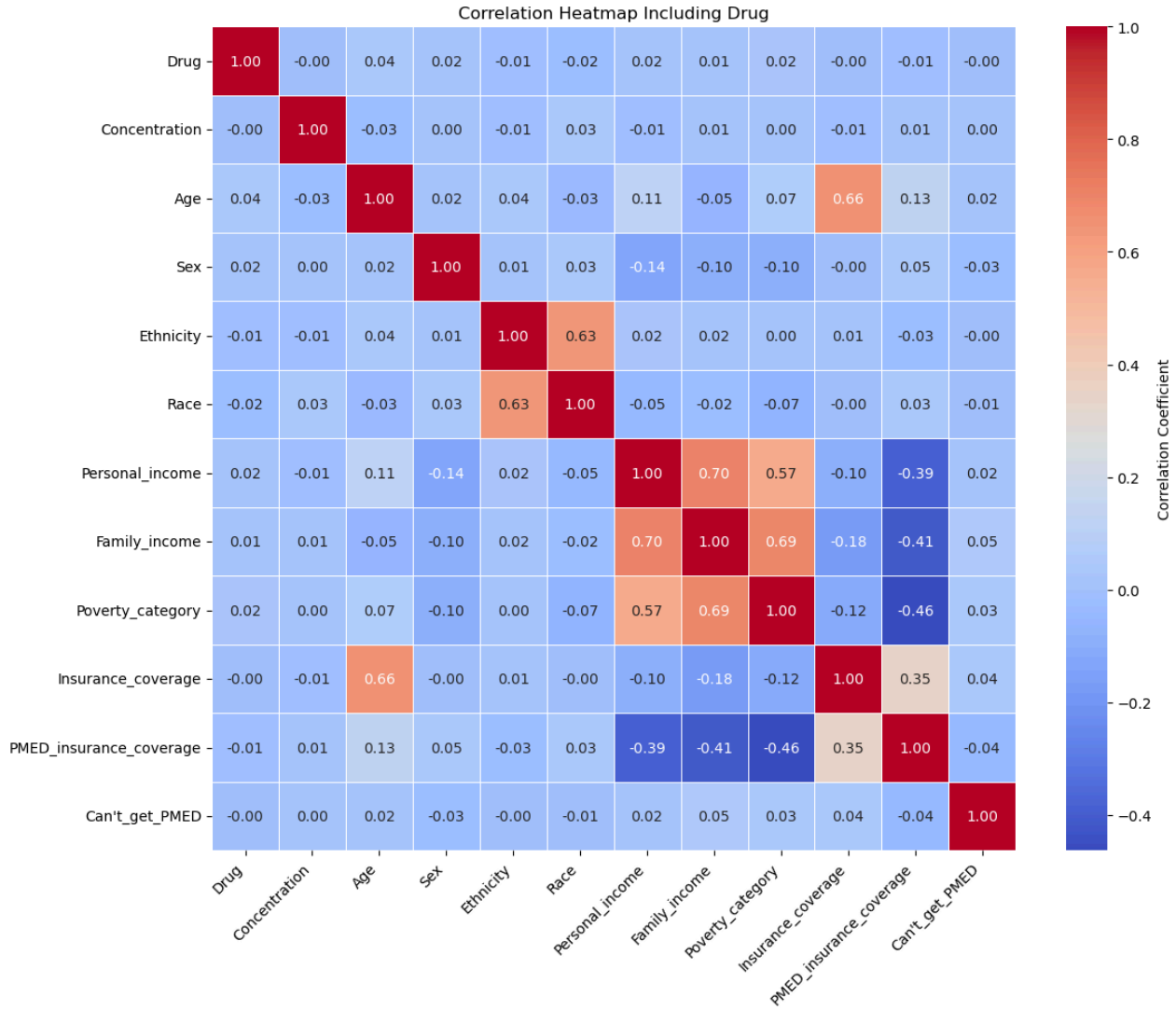
MultiOutput KNN model of Pharmasuticals Data of 2016

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsRegressor
from sklearn.multioutput import MultiOutputRegressor
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score, f1_score, confusion_matrix, recall_sco
from sklearn.preprocessing import StandardScaler, LabelEncoder
```

```
import numpy as np

# Load dataset
df = pd.read_csv("meps_clean_2016.csv")

# Define features (X) and target variables (y)
X = df[['Age', 'Sex', 'Ethnicity', 'Race', 'Personal_income', 'Family_income',
        'Poverty_category', 'Insurance_coverage', 'PMED_insurance_coverage', 'Can\'t_get_PMED']]
y = df[['Concentration', 'Drug']]

feature_set = ['Drug', 'Concentration', 'Age', 'Sex', 'Ethnicity',
               'Race', 'Personal_income', 'Family_income',
               'Poverty_category', 'Insurance_coverage',
               'PMED_insurance_coverage', 'Can\'t_get_PMED']

# Drop rows with missing values in any of the features or target variables
df_cleaned = df.dropna(subset=feature_set)

# Handle categorical columns in X (features) and convert them to numeric using LabelEncoder
categorical_columns = X.select_dtypes(include=['object']).columns
for col in categorical_columns:
    X[col] = LabelEncoder().fit_transform(X[col])

# Handle missing values in target variables y (Concentration and Drug)
encoder = LabelEncoder()
y['Drug'] = encoder.fit_transform(y['Drug'])

# Replace infinite values with NaN
y.replace([np.inf, -np.inf], np.nan, inplace=True)

# Convert Concentration to numeric values and handle any NaN values
y['Concentration'] = pd.to_numeric(y['Concentration'], errors='coerce')

# Ensure there are no missing values in features or targets after conversion
X = X.fillna(0)
y = y.fillna(0)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Standardize the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Initialize the KNN Regressor
knn_model = KNeighborsRegressor(n_neighbors=5)
multi_output_knn = MultiOutputRegressor(knn_model)

# Train the model
multi_output_knn.fit(X_train_scaled, y_train)

# Make predictions
y_pred = multi_output_knn.predict(X_test_scaled)

# Evaluate Concentration (regression)
mse = mean_squared_error(y_test['Concentration'], y_pred[:, 0])
r2 = r2_score(y_test['Concentration'], y_pred[:, 0])

# Print metrics for Concentration
print("\nRegression Metrics for Concentration:")
print(f"Mean Squared Error (MSE): {mse}")
print(f"R² Score: {r2}")

# Evaluate Drug (classification)
threshold = 0.5
y_test_binarized = (y_test['Drug'] >= threshold).astype(int)
y_pred_binarized = (y_pred[:, 1] >= threshold).astype(int)

conf_matrix = confusion_matrix(y_test_binarized, y_pred_binarized)
accuracy = accuracy_score(y_test_binarized, y_pred_binarized)
f1 = f1_score(y_test_binarized, y_pred_binarized)
recall = recall_score(y_test_binarized, y_pred_binarized)

# Print metrics for Drug
print("\nClassification Metrics for Drug:")
print(f"Accuracy: {accuracy}")
print(f"F1 Score: {f1}")
print(f"Recall: {recall}")
print("\nConfusion Matrix:")
print(conf_matrix)
```

```
# Create a DataFrame to compare actual vs predicted values
comparison_df = pd.DataFrame({
    'Actual_Concentration': y_test['Concentration'],
    'Predicted_Concentration': y_pred[:, 0],
    'Actual_Drug': y_test['Drug'],
    'Predicted_Drug': y_pred[:, 1]
})

# Add inverse-transformed Drug names
comparison_df['Predicted_Drug_Name'] = encoder.inverse_transform(comparison_df['Predicted_Drug'].round().astype(
comparison_df['Actual_Drug_Name'] = encoder.inverse_transform(comparison_df['Actual_Drug'].round().astype(int))

# Save results to CSV
comparison_df.to_csv("comparison_results_2016.csv", index=False)

# Display comparison table
print("\nComparison of Actual vs Predicted Values:")
print(comparison_df.head())

# Ensure 'Drug' is numeric and included in the cleaned dataset
df_cleaned['Drug'] = encoder.transform(df_cleaned['Drug'])

# Correlation matrix and heatmap (including Drug)
correlation_matrix = df_cleaned[feature_set].corr()

# Plot the heatmap
plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5,
            cbar_kws={'label': 'Correlation Coefficient'})
plt.title("Correlation Heatmap Including Drug")
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_327/1453418351.py:12: DtypeWarning: Columns (22) have mixed types. Specify dtype option on import
  df = pd.read_csv("meps_clean_2016.csv")

Regression Metrics for Concentration:
Mean Squared Error (MSE): 0.031499061769693186
R² Score: 0.14356985277355994

Classification Metrics for Drug:
Accuracy: 0.9938071780436313
F1 Score: 0.9968939714810109
Recall: 0.9999797685522376

Confusion Matrix:
[[    0   307]
 [    1 49427]]

Comparison of Actual vs Predicted Values:
        Actual_Concentration  Predicted_Concentration  Actual_Drug  \
88852           5.219005e-07                 0.000113            4
170870          8.054020e-05                 0.000202          185
224711          2.013505e-05                 0.000113          171
66285           2.965910e-04                 0.009779          235
223412          2.416206e-04                 0.000339          248

        Predicted_Drug Predicted_Drug_Name Actual_Drug_Name
88852             84.4   DEXTROAMPHETAMINE        ALBUTEROL
170870           180.4         LINACLOTIDE       LORATADINE
224711           116.4          FENOPROFEN        LETROZOLE
66285            206.2          METOPROLOL       OMEPRAZOLE
223412           158.4          IRBESARTAN      PHENTERMINE
```

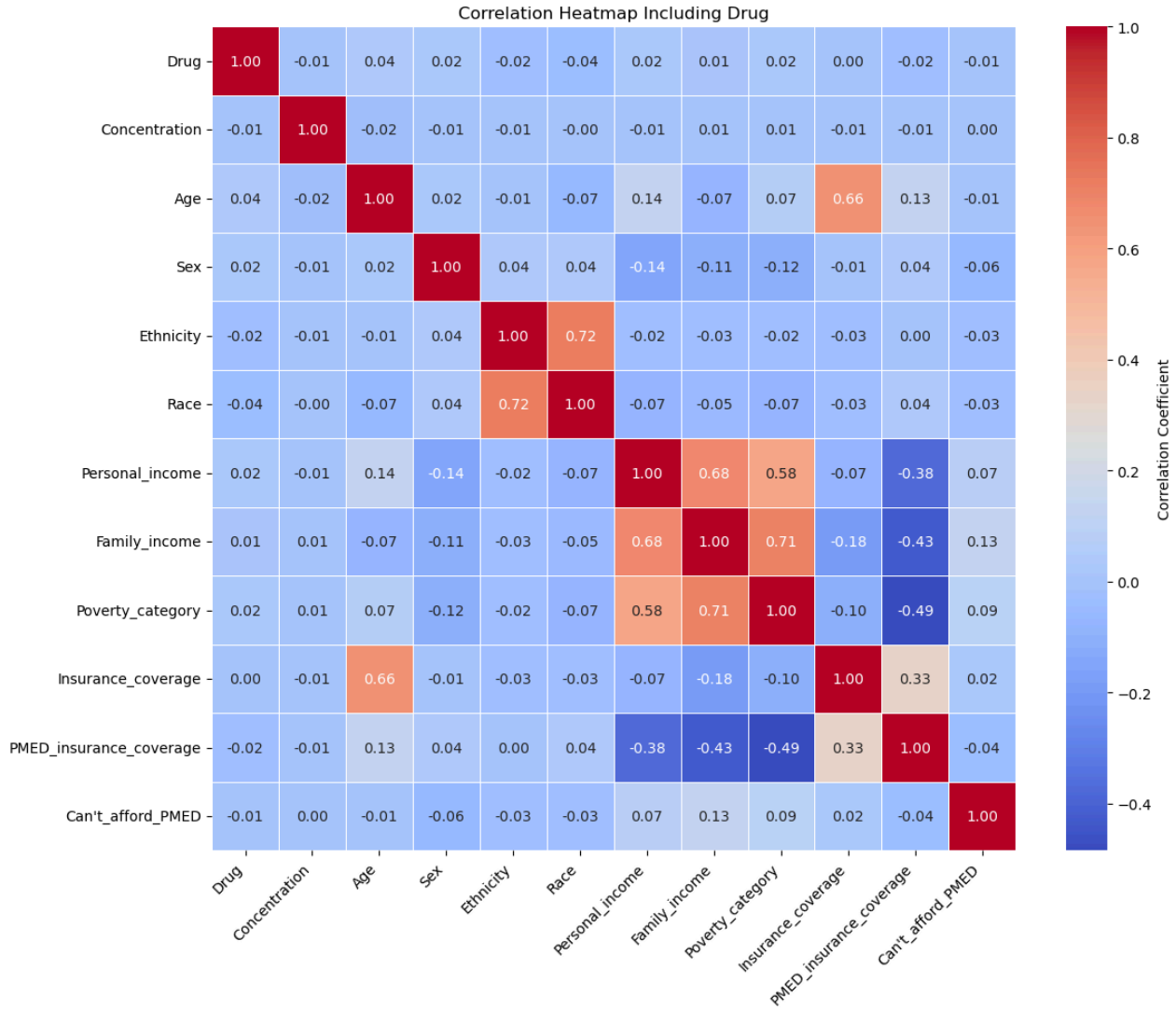


MultiOutput KNN-PCA model of Pharmasuticals Data of 2018

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsRegressor
from sklearn.multioutput import MultiOutputRegressor
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score, f1_score, confusion_matrix, recall_sco
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.decomposition import PCA
import numpy as np

# Load dataset
df = pd.read_csv("meps_clean_2018.csv")

# Define features (X) and target variables (y)
X = df[['Age', 'Sex', 'Ethnicity', 'Race', 'Personal_income', 'Family_income',
        'Poverty_category', 'Insurance_coverage', 'PMED_insurance_coverage', 'Can\'t_afford_PMED']]
y = df[['Concentration', 'Drug']]

feature_set = ['Drug', 'Concentration', 'Age', 'Sex', 'Ethnicity',
               'Race', 'Personal_income', 'Family_income',
               'Poverty_category', 'Insurance_coverage',
               'PMED_insurance_coverage', 'Can\'t_afford_PMED']

# Drop rows with missing values in any of the features or target variables
df_cleaned = df.dropna(subset=feature_set)

# Handle categorical columns in X (features) and convert them to numeric using LabelEncoder
categorical_columns = X.select_dtypes(include=['object']).columns
for col in categorical_columns:
    X[col] = LabelEncoder().fit_transform(X[col])

# Handle missing values in target variables y (Concentration and Drug)
encoder = LabelEncoder()
y['Drug'] = encoder.fit_transform(y['Drug'])

# Replace infinite values with NaN
y.replace([np.inf, -np.inf], np.nan, inplace=True)

# Convert Concentration to numeric values and handle any NaN values
y['Concentration'] = pd.to_numeric(y['Concentration'], errors='coerce')

# Ensure there are no missing values in features or targets after conversion
X = X.fillna(0)
y = y.fillna(0)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Standardize the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Apply PCA for dimensionality reduction
pca = PCA(n_components=2)  # Reduce to 2 components for visualization
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)

# Initialize the KNN Regressor
knn_model = KNeighborsRegressor(n_neighbors=8)
multi_output_knn = MultiOutputRegressor(knn_model)

# Train the model
multi_output_knn.fit(X_train_pca, y_train)

# Make predictions
y_pred = multi_output_knn.predict(X_test_pca)

# Evaluate Concentration (regression)
mse = mean_squared_error(y_test['Concentration'], y_pred[:, 0])
r2 = r2_score(y_test['Concentration'], y_pred[:, 0])

# Print metrics for Concentration
print("\nRegression Metrics for Concentration:")
print(f"Mean Squared Error (MSE): {mse}")
print(f"R² Score: {r2}")

# Evaluate Drug (classification)
```

```
threshold = 0.5
y_test_binarized = (y_test['Drug'] >= threshold).astype(int)
y_pred_binarized = (y_pred[:, 1] >= threshold).astype(int)

conf_matrix = confusion_matrix(y_test_binarized, y_pred_binarized)
accuracy = accuracy_score(y_test_binarized, y_pred_binarized)
f1 = f1_score(y_test_binarized, y_pred_binarized)
recall = recall_score(y_test_binarized, y_pred_binarized)

# Print metrics for Drug
print("\nClassification Metrics for Drug:")
print(f"Accuracy: {accuracy}")
print(f"F1 Score: {f1}")
print(f"Recall: {recall}")
print("\nConfusion Matrix:")
print(conf_matrix)

# Create a DataFrame to compare actual vs predicted values
comparison_df = pd.DataFrame({
    'Actual_Concentration': y_test['Concentration'],
    'Predicted_Concentration': y_pred[:, 0],
    'Actual_Drug': y_test['Drug'],
    'Predicted_Drug': y_pred[:, 1]
})

# Add inverse-transformed Drug names
comparison_df['Predicted_Drug_Name'] = encoder.inverse_transform(comparison_df['Predicted_Drug'].round().astype(
comparison_df['Actual_Drug_Name'] = encoder.inverse_transform(comparison_df['Actual_Drug'].round().astype(int))

# Save results to CSV
comparison_df.to_csv("comparison_results_2018.csv", index=False)

# Display comparison table
print("\nComparison of Actual vs Predicted Values:")
print(comparison_df.head())

# Ensure 'Drug' is numeric and included in the cleaned dataset
df_cleaned['Drug'] = encoder.transform(df_cleaned['Drug'])

# Correlation matrix and heatmap (including Drug)
correlation_matrix = df_cleaned[feature_set].corr()

# Plot the heatmap
plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5,
            cbar_kws={'label': 'Correlation Coefficient'})
plt.title("Correlation Heatmap Including Drug")
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_327/929319801.py:13: DtypeWarning: Columns (22) have mixed types. Specify dtype option on import o
  df = pd.read_csv("meps_clean_2018.csv")

Regression Metrics for Concentration:
Mean Squared Error (MSE): 0.4215067699020273
R² Score: 0.2178346722887352

Classification Metrics for Drug:
Accuracy: 0.9935188112551375
F1 Score: 0.9967486637801145
Recall: 1.0

Confusion Matrix:
[[    2   205]
 [    0 31423]]

Comparison of Actual vs Predicted Values:
        Actual_Concentration  Predicted_Concentration  Actual_Drug  \
131415              0.000016                 0.003899          227
125377              0.005234                 0.003250          135
54331               0.000322                 0.002502          184
110912              0.000161                 0.000375          134
44975               0.000805                 0.001440          188

        Predicted_Drug  Predicted_Drug_Name Actual_Drug_Name
131415         162.625         KETOCONAZOLE    NITROGLYCERIN
125377         213.375           MOMETASONE       GABAPENTIN
54331          143.625  HYDROCHLOROTHIAZIDE       LISINOPRIL
110912         178.125        LEVOTHYROXINE       FUROSEMIDE
44975          157.375          IPRATROPIUM         LOSARTAN
```

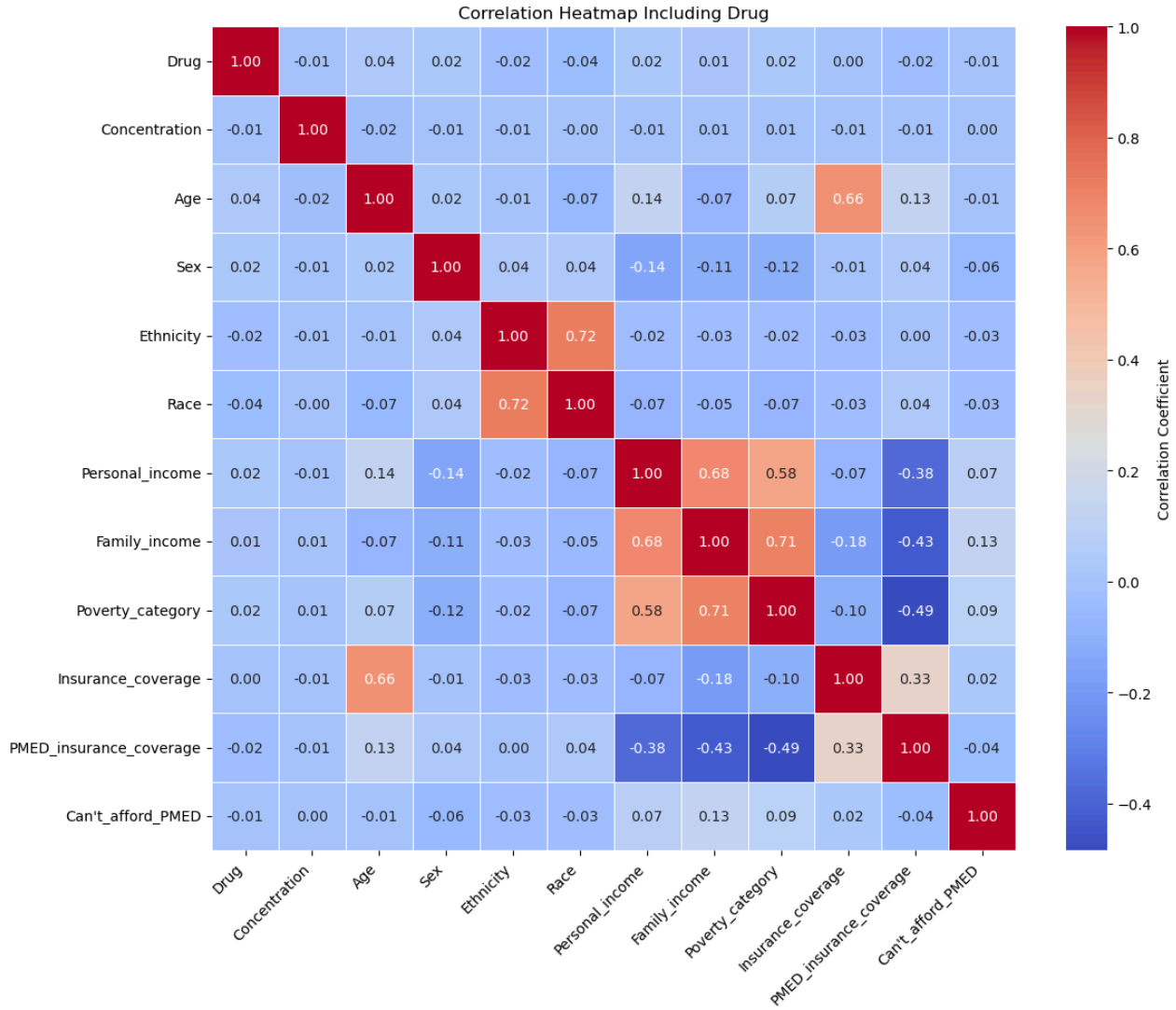


Test by reducing the Feature Set

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsRegressor
from sklearn.multioutput import MultiOutputRegressor
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score, f1_score, confusion_matrix, recall_sco
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.decomposition import PCA
import numpy as np

# Load dataset
df = pd.read_csv("meps_clean_2018.csv")

# Define features (X) and target variables (y)
X = df[['Family_income',
        'Poverty_category', 'Insurance_coverage', 'PMED_insurance_coverage', 'Can\'t_afford_PMED']]
y = df[['Concentration', 'Drug']]

feature_set = ['Drug', 'Concentration', 'Age', 'Sex', 'Ethnicity',
               'Race', 'Personal_income', 'Family_income',
               'Poverty_category', 'Insurance_coverage',
               'PMED_insurance_coverage', 'Can\'t_afford_PMED']

# Drop rows with missing values in any of the features or target variables
df_cleaned = df.dropna(subset=feature_set)

# Handle categorical columns in X (features) and convert them to numeric using LabelEncoder
categorical_columns = X.select_dtypes(include=['object']).columns
for col in categorical_columns:
    X[col] = LabelEncoder().fit_transform(X[col])

# Handle missing values in target variables y (Concentration and Drug)
encoder = LabelEncoder()
y['Drug'] = encoder.fit_transform(y['Drug'])

# Replace infinite values with NaN
y.replace([np.inf, -np.inf], np.nan, inplace=True)

# Convert Concentration to numeric values and handle any NaN values
y['Concentration'] = pd.to_numeric(y['Concentration'], errors='coerce')

# Ensure there are no missing values in features or targets after conversion
X = X.fillna(0)
y = y.fillna(0)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Standardize the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Apply PCA for dimensionality reduction
pca = PCA(n_components=2)  # Reduce to 2 components for visualization
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)

# Initialize the KNN Regressor
knn_model = KNeighborsRegressor(n_neighbors=8)
multi_output_knn = MultiOutputRegressor(knn_model)

# Train the model
multi_output_knn.fit(X_train_pca, y_train)

# Make predictions
y_pred = multi_output_knn.predict(X_test_pca)

# Evaluate Concentration (regression)
mse = mean_squared_error(y_test['Concentration'], y_pred[:, 0])
r2 = r2_score(y_test['Concentration'], y_pred[:, 0])

# Print metrics for Concentration
print("\nRegression Metrics for Concentration:")
print(f"Mean Squared Error (MSE): {mse}")
print(f"R² Score: {r2}")

# Evaluate Drug (classification)
```

```
threshold = 0.5
y_test_binarized = (y_test['Drug'] >= threshold).astype(int)
y_pred_binarized = (y_pred[:, 1] >= threshold).astype(int)

conf_matrix = confusion_matrix(y_test_binarized, y_pred_binarized)
accuracy = accuracy_score(y_test_binarized, y_pred_binarized)
f1 = f1_score(y_test_binarized, y_pred_binarized)
recall = recall_score(y_test_binarized, y_pred_binarized)

# Print metrics for Drug
print("\nClassification Metrics for Drug:")
print(f"Accuracy: {accuracy}")
print(f"F1 Score: {f1}")
print(f"Recall: {recall}")
print("\nConfusion Matrix:")
print(conf_matrix)

# Create a DataFrame to compare actual vs predicted values
comparison_df = pd.DataFrame({
    'Actual_Concentration': y_test['Concentration'],
    'Predicted_Concentration': y_pred[:, 0],
    'Actual_Drug': y_test['Drug'],
    'Predicted_Drug': y_pred[:, 1]
})

# Add inverse-transformed Drug names
comparison_df['Predicted_Drug_Name'] = encoder.inverse_transform(comparison_df['Predicted_Drug'].round().astype(
comparison_df['Actual_Drug_Name'] = encoder.inverse_transform(comparison_df['Actual_Drug'].round().astype(int))

# Save results to CSV
comparison_df.to_csv("comparison_results_2018.csv", index=False)

# Display comparison table
print("\nComparison of Actual vs Predicted Values:")
print(comparison_df.head())

# Ensure 'Drug' is numeric and included in the cleaned dataset
df_cleaned['Drug'] = encoder.transform(df_cleaned['Drug'])

# Correlation matrix and heatmap (including Drug)
correlation_matrix = df_cleaned[feature_set].corr()

# Plot the heatmap
plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5,
            cbar_kws={'label': 'Correlation Coefficient'})
plt.title("Correlation Heatmap Including Drug")
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_327/2691497001.py:13: DtypeWarning: Columns (22) have mixed types. Specify dtype option on import
  df = pd.read_csv("meps_clean_2018.csv")

Regression Metrics for Concentration:
Mean Squared Error (MSE): 0.4676301485349881
R² Score: 0.13224623067963392

Classification Metrics for Drug:
Accuracy: 0.9935188112551375
F1 Score: 0.9967486637801145
Recall: 1.0

Confusion Matrix:
[[    2   205]
 [    0 31423]]

Comparison of Actual vs Predicted Values:
        Actual_Concentration  Predicted_Concentration  Actual_Drug  \
131415              0.000016                 0.000420          227
125377              0.005234                 0.000424          135
54331               0.000322                 0.002270          184
110912              0.000161                 0.000015          134
44975               0.000805                 0.005056          188

        Predicted_Drug  Predicted_Drug_Name Actual_Drug_Name
131415         100.250             DOCUSATE    NITROGLYCERIN
125377         143.750  HYDROCHLOROTHIAZIDE       GABAPENTIN
54331          163.125         KETOCONAZOLE       LISINOPRIL
110912         208.375        METRONIDAZOLE       FUROSEMIDE
44975          134.750           GABAPENTIN         LOSARTAN
```

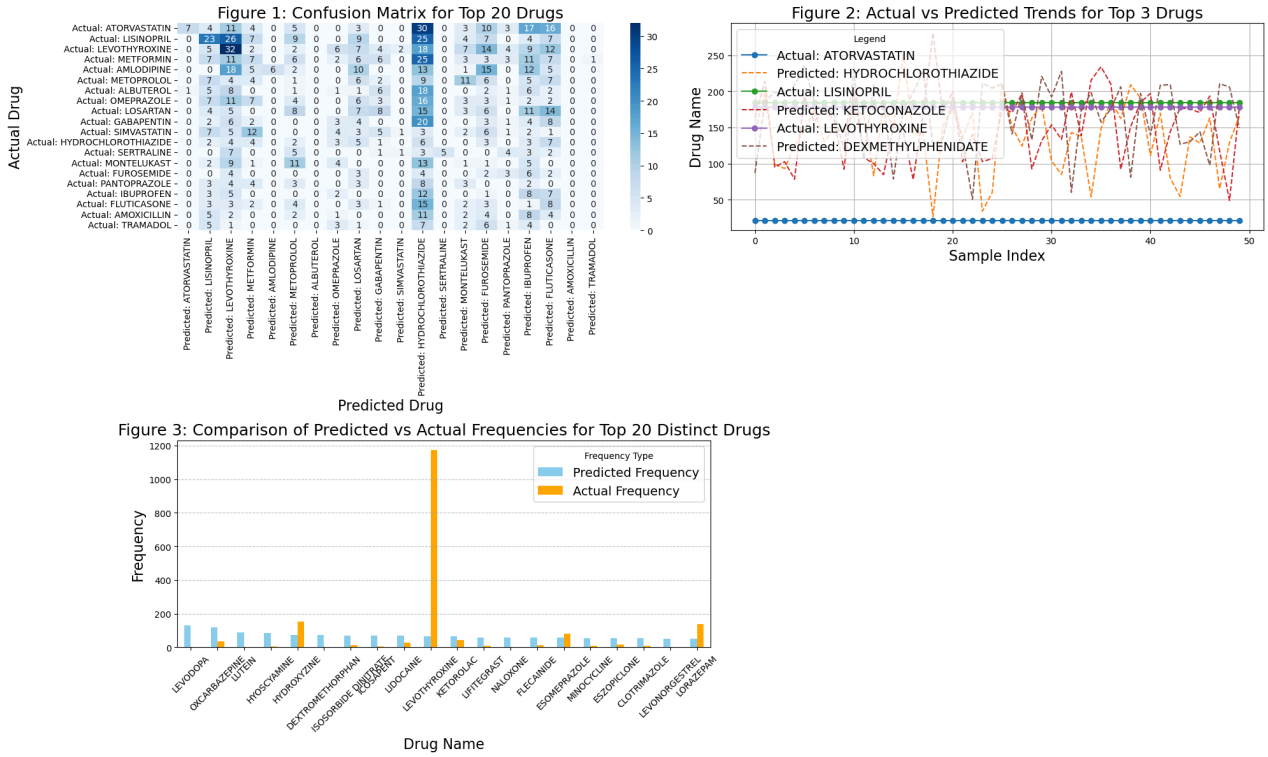


Analysis:

Reducing the similar feature sets the **R^2** value increased, which improved the model's efficiency in predicting concentration. However, there was no noticeable change in the accuracy of drug classification, indicating that the drug predictions remained stable despite adjustments to the features.

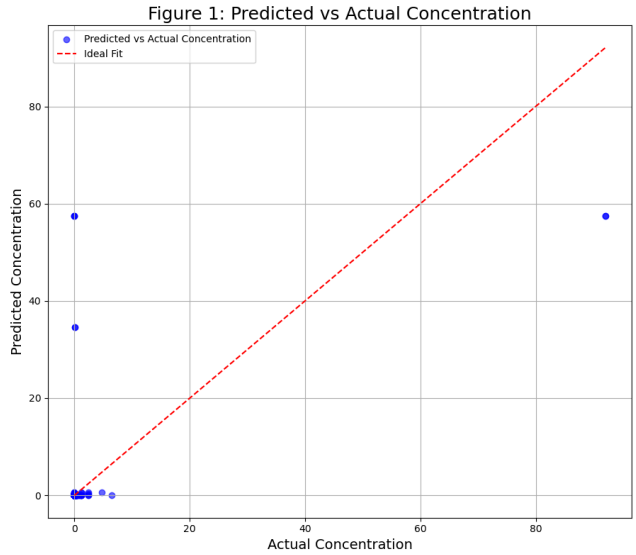
```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix
import numpy as np
import pandas as pd

# Assuming `df_cleaned`, `y_test`, `y_pred`, and `encoder` are already defined.

# Create a 2x2 grid for the plots, but only use 3 spaces with larger figsize
fig, axes = plt.subplots(2, 2, figsize=(24, 18))  # Increased the figsize to make all plots larger

top_n = 20  # You can change this to 10, 5, or any other number as required
top_drugs = df_cleaned['Drug'].value_counts().nlargest(top_n).index

# ---- Figure 1: Confusion Matrix for Top Drugs ----
filtered_y_test = y_test[y_test['Drug'].isin(top_drugs)]['Drug']
filtered_y_pred = y_pred[:, 1][np.isin(y_test['Drug'], top_drugs)].round()
conf_matrix = confusion_matrix(filtered_y_test, filtered_y_pred, labels=top_drugs)

sns.heatmap(
    conf_matrix,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=[f"Predicted: {encoder.inverse_transform([int(label)])[0]}" for label in top_drugs],
    yticklabels=[f"Actual: {encoder.inverse_transform([int(label)])[0]}" for label in top_drugs],
    ax=axes[0, 0]
)
axes[0, 0].set_title(f"Figure 1: Confusion Matrix for Top {top_n} Drugs", fontsize=18)
axes[0, 0].set_xlabel("Predicted Drug", fontsize=16)
axes[0, 0].set_ylabel("Actual Drug", fontsize=16)

# ---- Figure 2: Actual vs Predicted Trends for Top Drugs ----
for drug in top_drugs[:3]:  # Limiting to the top 5 for this graph
    actual_indices = y_test['Drug'] == drug
    actual = y_test.loc[actual_indices, 'Drug'].head(50).values
    predicted = y_pred[actual_indices, 1][:50]
    if len(actual) > 0:
        axes[0, 1].plot(
            range(len(actual)),
            actual,
            label=f"Actual: {encoder.inverse_transform([drug])[0]}",
            marker='o'
        )
        axes[0, 1].plot(
            range(len(predicted)),
            predicted,
            label=f"Predicted: {encoder.inverse_transform([int(predicted[0])])[0]}",
            linestyle='--'
        )
axes[0, 1].set_title(f"Figure 2: Actual vs Predicted Trends for Top 3 Drugs", fontsize=18)
axes[0, 1].set_xlabel("Sample Index", fontsize=16)
axes[0, 1].set_ylabel("Drug Name", fontsize=16)
axes[0, 1].legend(title="Legend", fontsize=14)
axes[0, 1].grid()

# ---- Figure 3: Bar Plot for Top Drugs ----
top_predicted_drugs = pd.Series(y_pred[:, 1]).value_counts().sort_values(ascending=False)
top_actual_frequencies = y_test['Drug'].value_counts()
common_drugs = top_predicted_drugs.index.intersection(top_actual_frequencies.index)

distinct_predicted_drugs = top_predicted_drugs.loc[common_drugs].head(top_n)
distinct_actual_frequencies = top_actual_frequencies.loc[common_drugs].head(top_n)

# Decode drug names using the LabelEncoder
decoded_drug_names = encoder.inverse_transform(distinct_predicted_drugs.index.astype(int))

# Create a comparison DataFrame for distinct drugs
comparison_df = pd.DataFrame({
    'Drug': decoded_drug_names,  # Use drug names for better readability
    'Predicted Frequency': distinct_predicted_drugs.values,
    'Actual Frequency': distinct_actual_frequencies.values
})
```

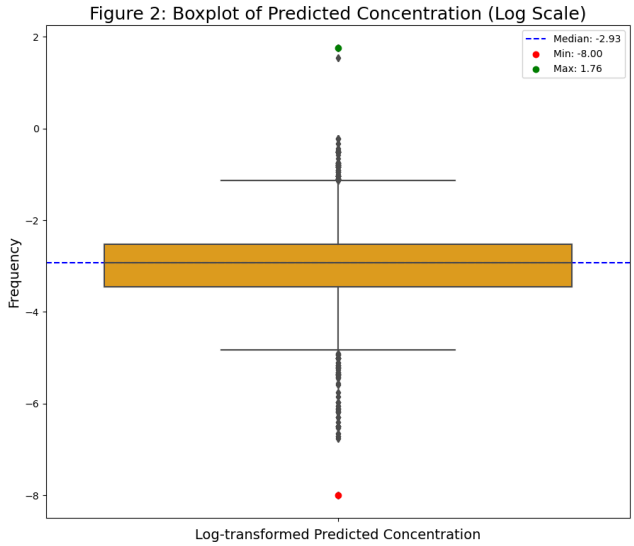
```
# Plot the comparison bar chart with a larger size
comparison_df.set_index('Drug').plot(kind='bar', figsize=(20, 12), rot=45, color=['skyblue', 'orange'], ax=axes[
axes[1, 0].set_title(f"Figure 3: Comparison of Predicted vs Actual Frequencies for Top {top_n} Distinct Drugs",
axes[1, 0].set_xlabel("Drug Name", fontsize=16)
axes[1, 0].set_ylabel("Frequency", fontsize=16)
axes[1, 0].legend(title="Frequency Type", loc='upper right', fontsize=14)
axes[1, 0].grid(axis='y', linestyle='--', alpha=0.7)

# Remove the extra plot (bottom-right corner) to leave only 3 plots
fig.delaxes(axes[1, 1])

# Adjust layout and show the plots
plt.tight_layout()
plt.show()
```




```
import numpy as np
import matplotlib.pyplot as plt
```

```
import seaborn as sns

fig, axs = plt.subplots(1, 2, figsize=(18, 8))

# --- Figure 1: Scatter plot, Predicted vs Actual Concentration ---

axs[0].scatter(y_test['Concentration'], y_pred[:, 0], alpha=0.6, color='blue', label='Predicted vs Actual Concen
axs[0].plot([min(y_test['Concentration']), max(y_test['Concentration'])],
            [min(y_test['Concentration']), max(y_test['Concentration'])],
            color='red', linestyle='--', label='Ideal Fit')

axs[0].set_title("Figure 1: Predicted vs Actual Concentration", fontsize=18)
axs[0].set_xlabel("Actual Concentration", fontsize=14)
axs[0].set_ylabel("Predicted Concentration", fontsize=14)
axs[0].legend()
axs[0].grid()


# --- Figure 2: Boxplot of Log-transformed Predicted Concentration ---

# Shift the values and apply log transformation to predicted concentration
shifted_concentration = y_pred[:, 0] - np.min(y_pred[:, 0]) + 1e-8
log_predicted_concentration = np.log10(shifted_concentration)

sns.boxplot(y=log_predicted_concentration, color='orange', ax=axs[1])

median_value = np.median(log_predicted_concentration)
axs[1].axhline(median_value, color='blue', linestyle='--', label=f'Median: {median_value:.2f}')
min_value = np.min(log_predicted_concentration)
max_value = np.max(log_predicted_concentration)
axs[1].scatter([0], [min_value], color='red', zorder=5, label=f'Min: {min_value:.2f}')
axs[1].scatter([0], [max_value], color='green', zorder=5, label=f'Max: {max_value:.2f}')
axs[1].set_title("Figure 2: Boxplot of Predicted Concentration (Log Scale)", fontsize=18)
axs[1].set_xlabel("Log-transformed Predicted Concentration", fontsize=14)
axs[1].set_ylabel("Frequency", fontsize=14)
axs[1].legend(loc="upper right")

plt.tight_layout()
plt.show()
```





Analysis:

For the **2018** data, the trends were similar to what we saw in **2014**. The drug predictions were mostly accurate, with solid performance in identifying commonly used medications. However, when it came to predicting concentrations, the results weren't as strong. There was a weak correlation between predicted and actual values, especially at higher concentrations, where the predictions often missed the mark. The model seems to rely more on demographic factors for concentration predictions, but there's definitely room for improvement, especially in making those concentration predictions more reliable.

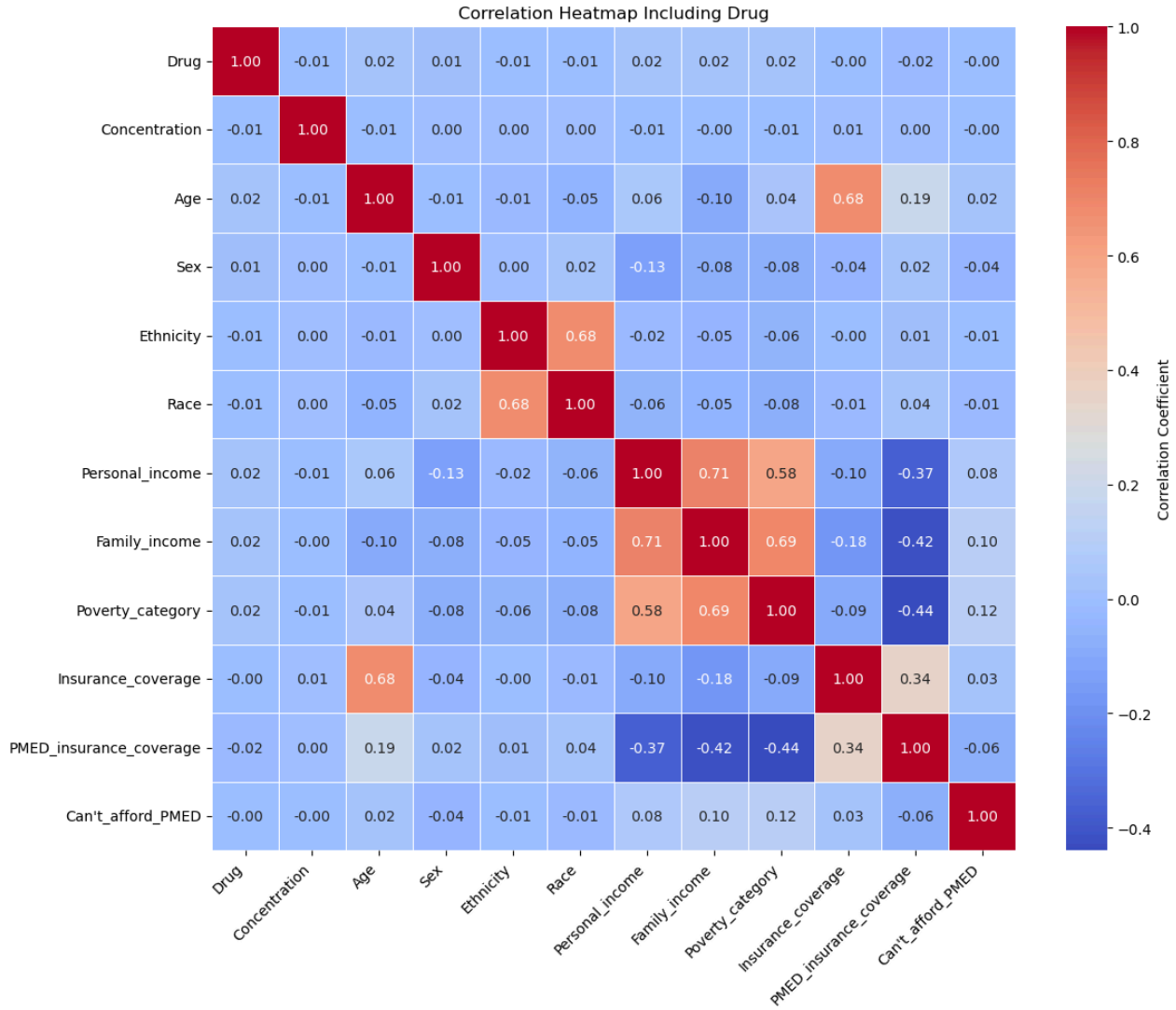
MultiOutput KNN-PCA Model for Pharmaceutical Data of 2020

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsRegressor
from sklearn.multioutput import MultiOutputRegressor
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score, f1_score, confusion_matrix, recall_sco
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.decomposition import PCA
import numpy as np

# Load dataset
df = pd.read_csv("meps_clean_2020.csv")

# Define features (X) and target variables (y)
X = df[['Age', 'Sex', 'Ethnicity', 'Race', 'Personal_income', 'Family_income',
        'Poverty_category', 'Insurance_coverage', 'PMED_insurance_coverage', 'Can\'t_afford_PMED']]
y = df[['Concentration', 'Drug']]

feature_set = ['Drug', 'Concentration', 'Age', 'Sex', 'Ethnicity',
               'Race', 'Personal_income', 'Family_income',
               'Poverty_category', 'Insurance_coverage',
               'PMED_insurance_coverage', 'Can\'t_afford_PMED']

# Drop rows with missing values in any of the features or target variables
df_cleaned = df.dropna(subset=feature_set)
```

```
# Handle categorical columns in X (features) and convert them to numeric using LabelEncoder
categorical_columns = X.select_dtypes(include=['object']).columns
for col in categorical_columns:
    X[col] = LabelEncoder().fit_transform(X[col])

# Handle missing values in target variables y (Concentration and Drug)
encoder = LabelEncoder()
y['Drug'] = encoder.fit_transform(y['Drug'])

# Replace infinite values with NaN
y.replace([np.inf, -np.inf], np.nan, inplace=True)

# Convert Concentration to numeric values and handle any NaN values
y['Concentration'] = pd.to_numeric(y['Concentration'], errors='coerce')

# Ensure there are no missing values in features or targets after conversion
X = X.fillna(0)
y = y.fillna(0)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Standardize the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Apply PCA for dimensionality reduction
pca = PCA(n_components=2)  # Reduce to 2 components for visualization
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)

# Initialize the KNN Regressor
knn_model = KNeighborsRegressor(n_neighbors=8)
multi_output_knn = MultiOutputRegressor(knn_model)

# Train the model
multi_output_knn.fit(X_train_pca, y_train)

# Make predictions
y_pred = multi_output_knn.predict(X_test_pca)

# Evaluate Concentration (regression)
mse = mean_squared_error(y_test['Concentration'], y_pred[:, 0])
r2 = r2_score(y_test['Concentration'], y_pred[:, 0])

# Print metrics for Concentration
print("\nRegression Metrics for Concentration:")
print(f"Mean Squared Error (MSE): {mse}")
print(f"R² Score: {r2}")

# Evaluate Drug (classification)
threshold = 0.5
y_test_binarized = (y_test['Drug'] >= threshold).astype(int)
y_pred_binarized = (y_pred[:, 1] >= threshold).astype(int)

conf_matrix = confusion_matrix(y_test_binarized, y_pred_binarized)
accuracy = accuracy_score(y_test_binarized, y_pred_binarized)
f1 = f1_score(y_test_binarized, y_pred_binarized)
recall = recall_score(y_test_binarized, y_pred_binarized)

# Print metrics for Drug
print("\nClassification Metrics for Drug:")
print(f"Accuracy: {accuracy}")
print(f"F1 Score: {f1}")
print(f"Recall: {recall}")
print("\nConfusion Matrix:")
print(conf_matrix)

# Create a DataFrame to compare actual vs predicted values
comparison_df = pd.DataFrame({
    'Actual_Concentration': y_test['Concentration'],
    'Predicted_Concentration': y_pred[:, 0],
    'Actual_Drug': y_test['Drug'],
    'Predicted_Drug': y_pred[:, 1]
})

# Add inverse-transformed Drug names
comparison_df['Predicted_Drug_Name'] = encoder.inverse_transform(comparison_df['Predicted_Drug'].round().astype(
comparison_df['Actual_Drug_Name'] = encoder.inverse_transform(comparison_df['Actual_Drug'].round().astype(int))
```

```
# Save results to CSV
comparison_df.to_csv("comparison_results_2020.csv", index=False)

# Display comparison table
print("\nComparison of Actual vs Predicted Values:")
print(comparison_df.head())

# Ensure 'Drug' is numeric and included in the cleaned dataset
df_cleaned['Drug'] = encoder.transform(df_cleaned['Drug'])

# Correlation matrix and heatmap (including Drug)
correlation_matrix = df_cleaned[feature_set].corr()

# Plot the heatmap
plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5,
            cbar_kws={'label': 'Correlation Coefficient'})
plt.title("Correlation Heatmap Including Drug")
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_327/2239775387.py:13: DtypeWarning: Columns (20) have mixed types. Specify dtype option on import
  df = pd.read_csv("meps_clean_2020.csv")

Regression Metrics for Concentration:
Mean Squared Error (MSE): 0.005299623424217472
R² Score: 0.029500401017701683

Classification Metrics for Drug:
Accuracy: 0.9964254617111956
F1 Score: 0.998209530805252
Recall: 1.0

Confusion Matrix:
[[    0   156]
 [    0 43486]]

Comparison of Actual vs Predicted Values:
        Actual_Concentration  Predicted_Concentration  Actual_Drug  \
44725               0.000040                 0.000086          173
187359              0.000003                 0.000178          213
179514              0.000488                 0.001886           19
179371              0.000201                 0.001072           19
112089              0.000322                 0.000185          139

        Predicted_Drug Predicted_Drug_Name Actual_Drug_Name
44725          190.625     METHYLPHENIDATE       LISINOPRIL
187359         159.875         LATANOPROST    NITROGLYCERIN
179514         247.750     PSEUDOEPHEDRINE         ATENOLOL
179371         139.000         HYDROXYZINE         ATENOLOL
112089         143.625           IMIQUIMOD      HYDROXYZINE
```

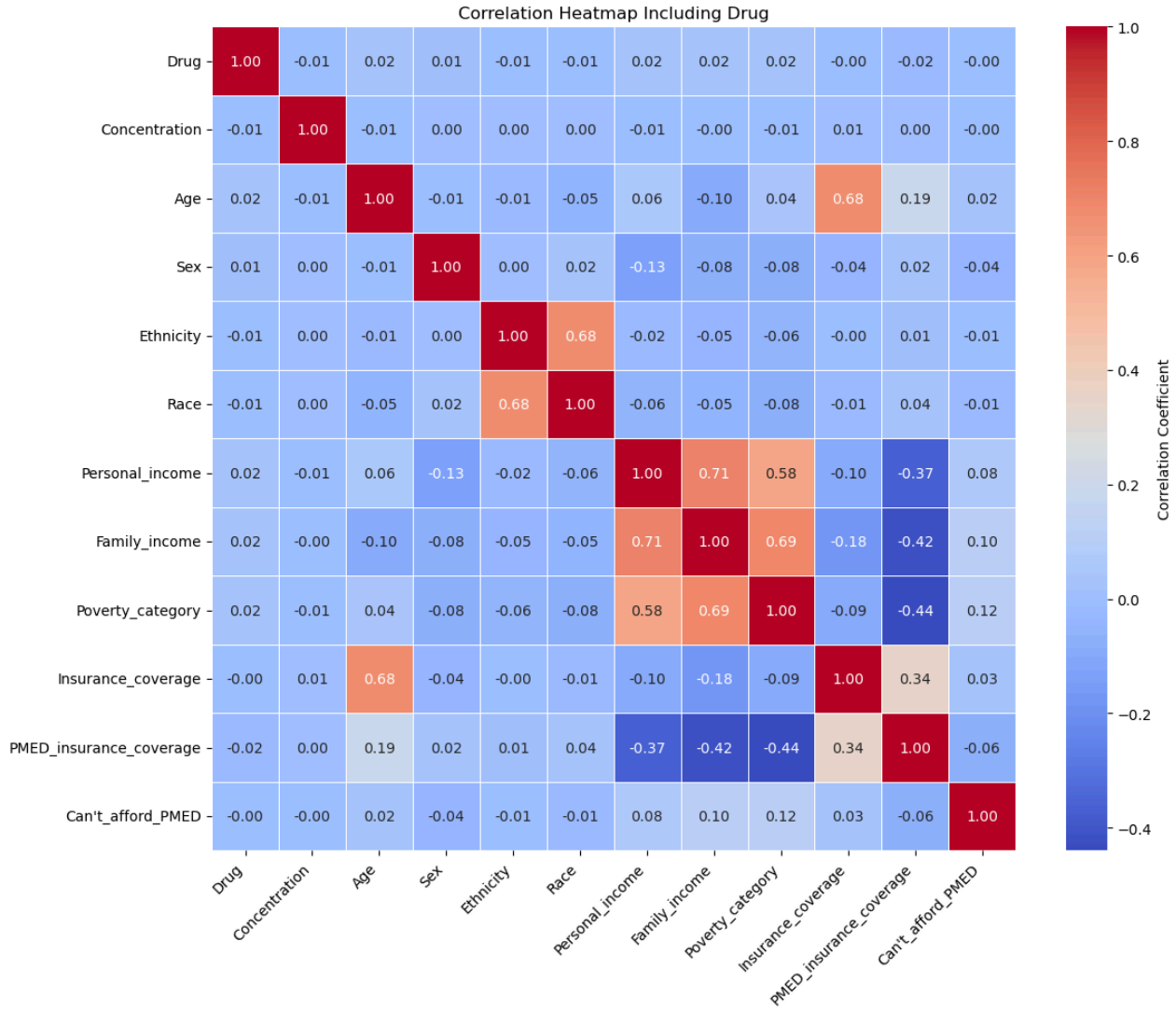


Reducing the features

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsRegressor
from sklearn.multioutput import MultiOutputRegressor
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score, f1_score, confusion_matrix, recall_sco
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.decomposition import PCA
import numpy as np

# Load dataset
df = pd.read_csv("meps_clean_2020.csv")

# Define features (X) and target variables (y)
X = df[['Age', 'Sex', 'Ethnicity', 'Personal_income',
        'Poverty_category', 'Insurance_coverage', 'PMED_insurance_coverage', 'Can\'t_afford_PMED']]
y = df[['Concentration', 'Drug']]

feature_set = ['Drug', 'Concentration', 'Age', 'Sex', 'Ethnicity',
               'Race', 'Personal_income', 'Family_income',
               'Poverty_category', 'Insurance_coverage',
               'PMED_insurance_coverage', 'Can\'t_afford_PMED']

# Drop rows with missing values in any of the features or target variables
df_cleaned = df.dropna(subset=feature_set)

# Handle categorical columns in X (features) and convert them to numeric using LabelEncoder
categorical_columns = X.select_dtypes(include=['object']).columns
for col in categorical_columns:
    X[col] = LabelEncoder().fit_transform(X[col])

# Handle missing values in target variables y (Concentration and Drug)
encoder = LabelEncoder()
y['Drug'] = encoder.fit_transform(y['Drug'])

# Replace infinite values with NaN
y.replace([np.inf, -np.inf], np.nan, inplace=True)

# Convert Concentration to numeric values and handle any NaN values
y['Concentration'] = pd.to_numeric(y['Concentration'], errors='coerce')

# Ensure there are no missing values in features or targets after conversion
X = X.fillna(0)
y = y.fillna(0)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Standardize the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Apply PCA for dimensionality reduction
pca = PCA(n_components=2)  # Reduce to 2 components for visualization
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)

# Initialize the KNN Regressor
knn_model = KNeighborsRegressor(n_neighbors=8)
multi_output_knn = MultiOutputRegressor(knn_model)

# Train the model
multi_output_knn.fit(X_train_pca, y_train)

# Make predictions
y_pred = multi_output_knn.predict(X_test_pca)

# Evaluate Concentration (regression)
mse = mean_squared_error(y_test['Concentration'], y_pred[:, 0])
r2 = r2_score(y_test['Concentration'], y_pred[:, 0])

# Print metrics for Concentration
print("\nRegression Metrics for Concentration:")
print(f"Mean Squared Error (MSE): {mse}")
print(f"R² Score: {r2}")

# Evaluate Drug (classification)
```

```
threshold = 0.5
y_test_binarized = (y_test['Drug'] >= threshold).astype(int)
y_pred_binarized = (y_pred[:, 1] >= threshold).astype(int)

conf_matrix = confusion_matrix(y_test_binarized, y_pred_binarized)
accuracy = accuracy_score(y_test_binarized, y_pred_binarized)
f1 = f1_score(y_test_binarized, y_pred_binarized)
recall = recall_score(y_test_binarized, y_pred_binarized)

# Print metrics for Drug
print("\nClassification Metrics for Drug:")
print(f"Accuracy: {accuracy}")
print(f"F1 Score: {f1}")
print(f"Recall: {recall}")
print("\nConfusion Matrix:")
print(conf_matrix)

# Create a DataFrame to compare actual vs predicted values
comparison_df = pd.DataFrame({
    'Actual_Concentration': y_test['Concentration'],
    'Predicted_Concentration': y_pred[:, 0],
    'Actual_Drug': y_test['Drug'],
    'Predicted_Drug': y_pred[:, 1]
})

# Add inverse-transformed Drug names
comparison_df['Predicted_Drug_Name'] = encoder.inverse_transform(comparison_df['Predicted_Drug'].round().astype(
comparison_df['Actual_Drug_Name'] = encoder.inverse_transform(comparison_df['Actual_Drug'].round().astype(int))

# Save results to CSV
comparison_df.to_csv("comparison_results_2020.csv", index=False)

# Display comparison table
print("\nComparison of Actual vs Predicted Values:")
print(comparison_df.head())

# Ensure 'Drug' is numeric and included in the cleaned dataset
df_cleaned['Drug'] = encoder.transform(df_cleaned['Drug'])

# Correlation matrix and heatmap (including Drug)
correlation_matrix = df_cleaned[feature_set].corr()

# Plot the heatmap
plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5,
            cbar_kws={'label': 'Correlation Coefficient'})
plt.title("Correlation Heatmap Including Drug")
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_327/3564563746.py:13: DtypeWarning: Columns (20) have mixed types. Specify dtype option on import
  df = pd.read_csv("meps_clean_2020.csv")

Regression Metrics for Concentration:
Mean Squared Error (MSE): 0.005387859647816568
R² Score: 0.013342041684598738

Classification Metrics for Drug:
Accuracy: 0.9964254617111956
F1 Score: 0.998209530805252
Recall: 1.0

Confusion Matrix:
[[    0   156]
 [    0 43486]]

Comparison of Actual vs Predicted Values:
        Actual_Concentration  Predicted_Concentration  Actual_Drug  \
44725               0.000040                 0.000116          173
187359              0.000003                 0.000340          213
179514              0.000488                 0.005258           19
179371              0.000201                 0.002099           19
112089              0.000322                 0.000186          139

        Predicted_Drug Predicted_Drug_Name Actual_Drug_Name
44725          202.375            MORPHINE       LISINOPRIL
187359         189.000       METHOCARBAMOL    NITROGLYCERIN
179514         168.625           LIDOCAINE         ATENOLOL
179371         184.375           MEMANTINE         ATENOLOL
112089         160.875         LEFLUNOMIDE      HYDROXYZINE
```

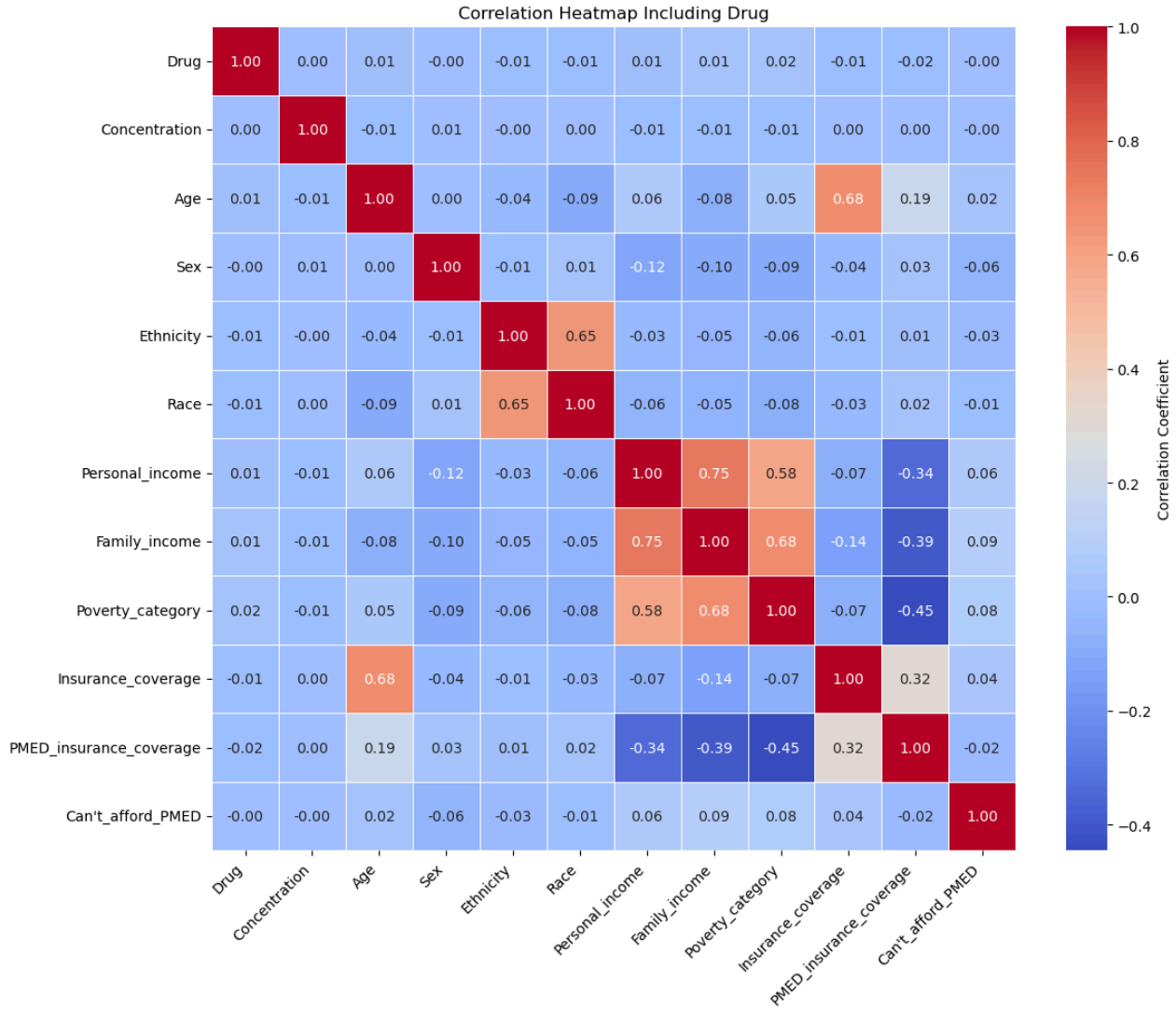


MultiOutput KNN-PCA Model for Pharmaceutical Data of 2021

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsRegressor
from sklearn.multioutput import MultiOutputRegressor
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score, f1_score, confusion_matrix, recall_sco
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.decomposition import PCA
import numpy as np

# Load dataset
df = pd.read_csv("meps_clean_2021.csv")

# Define features (X) and target variables (y)
X = df[['Age', 'Sex', 'Ethnicity', 'Race', 'Personal_income', 'Family_income',
        'Poverty_category', 'Insurance_coverage', 'PMED_insurance_coverage', 'Can\'t_afford_PMED']]
y = df[['Concentration', 'Drug']]

feature_set = ['Drug', 'Concentration', 'Age', 'Sex', 'Ethnicity',
               'Race', 'Personal_income', 'Family_income',
               'Poverty_category', 'Insurance_coverage',
               'PMED_insurance_coverage', 'Can\'t_afford_PMED']

# Drop rows with missing values in any of the features or target variables
df_cleaned = df.dropna(subset=feature_set)

# Handle categorical columns in X (features) and convert them to numeric using LabelEncoder
categorical_columns = X.select_dtypes(include=['object']).columns
for col in categorical_columns:
    X[col] = LabelEncoder().fit_transform(X[col])

# Handle missing values in target variables y (Concentration and Drug)
encoder = LabelEncoder()
y['Drug'] = encoder.fit_transform(y['Drug'])

# Replace infinite values with NaN
y.replace([np.inf, -np.inf], np.nan, inplace=True)

# Convert Concentration to numeric values and handle any NaN values
y['Concentration'] = pd.to_numeric(y['Concentration'], errors='coerce')

# Ensure there are no missing values in features or targets after conversion
X = X.fillna(0)
y = y.fillna(0)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Standardize the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Apply PCA for dimensionality reduction
pca = PCA(n_components=2)  # Reduce to 2 components for visualization
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)

# Initialize the KNN Regressor
knn_model = KNeighborsRegressor(n_neighbors=9)
multi_output_knn = MultiOutputRegressor(knn_model)

# Train the model
multi_output_knn.fit(X_train_pca, y_train)

# Make predictions
y_pred = multi_output_knn.predict(X_test_pca)

# Evaluate Concentration (regression)
mse = mean_squared_error(y_test['Concentration'], y_pred[:, 0])
r2 = r2_score(y_test['Concentration'], y_pred[:, 0])

# Print metrics for Concentration
print("\nRegression Metrics for Concentration:")
print(f"Mean Squared Error (MSE): {mse}")
print(f"R² Score: {r2}")

# Evaluate Drug (classification)
```

```
threshold = 0.5
y_test_binarized = (y_test['Drug'] >= threshold).astype(int)
y_pred_binarized = (y_pred[:, 1] >= threshold).astype(int)

conf_matrix = confusion_matrix(y_test_binarized, y_pred_binarized)
accuracy = accuracy_score(y_test_binarized, y_pred_binarized)
f1 = f1_score(y_test_binarized, y_pred_binarized)
recall = recall_score(y_test_binarized, y_pred_binarized)

# Print metrics for Drug
print("\nClassification Metrics for Drug:")
print(f"Accuracy: {accuracy}")
print(f"F1 Score: {f1}")
print(f"Recall: {recall}")
print("\nConfusion Matrix:")
print(conf_matrix)

# Create a DataFrame to compare actual vs predicted values
comparison_df = pd.DataFrame({
    'Actual_Concentration': y_test['Concentration'],
    'Predicted_Concentration': y_pred[:, 0],
    'Actual_Drug': y_test['Drug'],
    'Predicted_Drug': y_pred[:, 1]
})

# Add inverse-transformed Drug names
comparison_df['Predicted_Drug_Name'] = encoder.inverse_transform(comparison_df['Predicted_Drug'].round().astype(
comparison_df['Actual_Drug_Name'] = encoder.inverse_transform(comparison_df['Actual_Drug'].round().astype(int))

# Save results to CSV
comparison_df.to_csv("comparison_results_2021.csv", index=False)

# Display comparison table
print("\nComparison of Actual vs Predicted Values:")
print(comparison_df.head())

# Ensure 'Drug' is numeric and included in the cleaned dataset
df_cleaned['Drug'] = encoder.transform(df_cleaned['Drug'])

# Correlation matrix and heatmap (including Drug)
correlation_matrix = df_cleaned[feature_set].corr()

# Plot the heatmap
plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5,
            cbar_kws={'label': 'Correlation Coefficient'})
plt.title("Correlation Heatmap Including Drug")
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_327/1319645556.py:13: DtypeWarning: Columns (20) have mixed types. Specify dtype option on import
  df = pd.read_csv("meps_clean_2021.csv")

Regression Metrics for Concentration:
Mean Squared Error (MSE): 0.028549544405492736
R² Score: 0.07971858888233985

Classification Metrics for Drug:
Accuracy: 0.9958830614255995
F1 Score: 0.9979372846766584
Recall: 1.0

Confusion Matrix:
[[    0   188]
 [    0 45477]]

Comparison of Actual vs Predicted Values:
        Actual_Concentration  Predicted_Concentration  Actual_Drug  \
58514               0.000040                 0.000121           19
134578              0.000805                 0.004550          151
87135               0.000004                 0.000298          242
142646              0.000081                 0.000335           82
228053              0.000322                 0.003268          128

        Predicted_Drug Predicted_Drug_Name Actual_Drug_Name
58514        97.666667        FLURBIPROFEN     ATORVASTATIN
134578      138.444444           LETROZOLE         LOSARTAN
87135       137.666667           LETROZOLE       TAMSULOSIN
142646       95.777778        FLUOROURACIL     ESCITALOPRAM
228053      171.000000         MONTELUKAST     ISOTRETINOIN
```



Reducing features

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsRegressor
from sklearn.multioutput import MultiOutputRegressor
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score, f1_score, confusion_matrix, recall_sco
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.decomposition import PCA
import numpy as np

# Load dataset
df = pd.read_csv("meps_clean_2021.csv")

# Define features (X) and target variables (y)
X = df[['Age', 'Sex', 'Race', 'Family_income',
        'Poverty_category', 'Insurance_coverage', 'PMED_insurance_coverage', 'Can\'t_afford_PMED']]
y = df[['Concentration', 'Drug']]

feature_set = ['Drug', 'Concentration', 'Age', 'Sex', 'Ethnicity',
               'Race', 'Personal_income', 'Family_income',
               'Poverty_category', 'Insurance_coverage',
               'PMED_insurance_coverage', 'Can\'t_afford_PMED']

# Drop rows with missing values in any of the features or target variables
df_cleaned = df.dropna(subset=feature_set)

# Handle categorical columns in X (features) and convert them to numeric using LabelEncoder
categorical_columns = X.select_dtypes(include=['object']).columns
for col in categorical_columns:
    X[col] = LabelEncoder().fit_transform(X[col])

# Handle missing values in target variables y (Concentration and Drug)
encoder = LabelEncoder()
y['Drug'] = encoder.fit_transform(y['Drug'])

# Replace infinite values with NaN
y.replace([np.inf, -np.inf], np.nan, inplace=True)

# Convert Concentration to numeric values and handle any NaN values
y['Concentration'] = pd.to_numeric(y['Concentration'], errors='coerce')

# Ensure there are no missing values in features or targets after conversion
X = X.fillna(0)
y = y.fillna(0)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Standardize the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Apply PCA for dimensionality reduction
pca = PCA(n_components=2)  # Reduce to 2 components for visualization
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)

# Initialize the KNN Regressor
knn_model = KNeighborsRegressor(n_neighbors=9)
multi_output_knn = MultiOutputRegressor(knn_model)

# Train the model
multi_output_knn.fit(X_train_pca, y_train)

# Make predictions
y_pred = multi_output_knn.predict(X_test_pca)

# Evaluate Concentration (regression)
mse = mean_squared_error(y_test['Concentration'], y_pred[:, 0])
r2 = r2_score(y_test['Concentration'], y_pred[:, 0])

# Print metrics for Concentration
print("\nRegression Metrics for Concentration:")
print(f"Mean Squared Error (MSE): {mse}")
print(f"R² Score: {r2}")

# Evaluate Drug (classification)
```